

From what I know so far:

-  **Propensity scores are shared** across the X-learner, Causal Forest, and DR Learner.
-  **Outcome model predictions are probably *not* shared.**

Also, there is a causal forest in parenthesis where the Model Specification is defined. It isn't a causal forest, just a random forest. Causal forest there is misleading and wrong.

Model Specification

The doubly robust learner is implemented using `econml.dr.ForestDRLearner`, which constructs doubly robust pseudo-outcomes and fits a final-stage causal forest to predict heterogeneous treatment effects.

The model combines two nuisance models with a final forest estimator:

Component	Model	Purpose
Outcome model	<code>RandomForestRegressor</code>	Estimates baseline ED risk (μ_0, μ_1)
Treatment model	<code>LogisticRegressionCV</code> (elastic net)	Estimates treatment propensity ($\hat{e}$)
Final stage	<code>ForestDRLearner</code> (causal forest)	Regresses doubly robust pseudo-outcomes on features to predict $\tau(x)$

The doubly robust pseudo-outcome for each member is constructed as:

$$\tilde{Y}_{DR} = \mu_1(x) - \mu_0(x) + W/\hat{e}(x) * (Y - \mu_1(x)) - (1-W)/(1-\hat{e}(x)) * (Y - \mu_0(x))$$



The mermaid diagram is not good at all, need to rework it to be more accurate rather than trying to copy the causal forest exactly. _ I want this process to be clear:

For the DR learner. After we get outcomes, we do what the t learner did and subtract the predicted outcomes from each other for treatment effect. However we also have a 2nd and third term. Each term is a correction factor. If the person was really treated, we take their treated pred and get the difference from their true outcome. Then this is weighted by the inverse of propensity. We do the same for the untreated people as well. Then we have a pseudo outcome as the sum of all three terms. We have a final model on top of the pseudo outcome to predict treatment effect

The propensity and overlap checks are already present in the causal forest ipynb, so no need to show here again...

Instead of showing the Doubly Robust Training Set Effect Distirbution in task 3, I should show the Test Set Effect Distribution only in task 4. No need for the training set one in task 3.

This visual is not the same format as the one in the casual forest. I want the same format please. You may need to refer to the generate readme tables.py to see how the table was done.

Risk Tier Versus Benefit Group

The chart below compares baseline risk tier against model-relative doubly robust benefit group. Benefit groups are based on HTE deciles rather than fixed absolute ED-risk-reduction thresholds, because the observed estimated benefits in this dataset are smaller than the idealized synthetic-data specification.

Benefit group	Definition
High benefit	HTE deciles 1-2, top 20% by estimated benefit
Medium benefit	HTE deciles 3-7, middle 50% by estimated benefit
Low benefit	HTE deciles 8-10, bottom 30% by estimated benefit

Risk tiers are based on `current_risk_score` : Low `<35` , Medium `35 to <55` , High `55 to <75` , and Very High `>=75` .

